IOT SECURITY & CO-SIMULATION FRAMEWORK

ERW-IoT: Real-Time Wormhole Attack Detection Engine

A Lightweight Path Consistency Validation Architecture Co-Simulated with NI LabVIEW Dashboard Host and Embedded MATLAB Engine

NI LabVIEW Dashboard | MATLAB Analysis Engine | ERW-IoT Validation

Project Aim & Objectives

- **ERW-IoT Engine Integration:** Introduce ERW-IoT to provide real-time wormhole detection via lightweight path consistency checks without heavy cryptographic overhead.
- **Wormhole Attack Mitigation:** Identify deceptive shortcut routes created by covert attacker tunnels linking geographically distant network nodes.
- **Physical Consistency Bounds:** Evaluate hop count vs transmission latency bounds to determine if proposed routes make physical sense.
- **Closed-Loop Operator Dashboard:** Provide immediate visual alarm notifications and topology map updates for human operators upon anomaly detection.

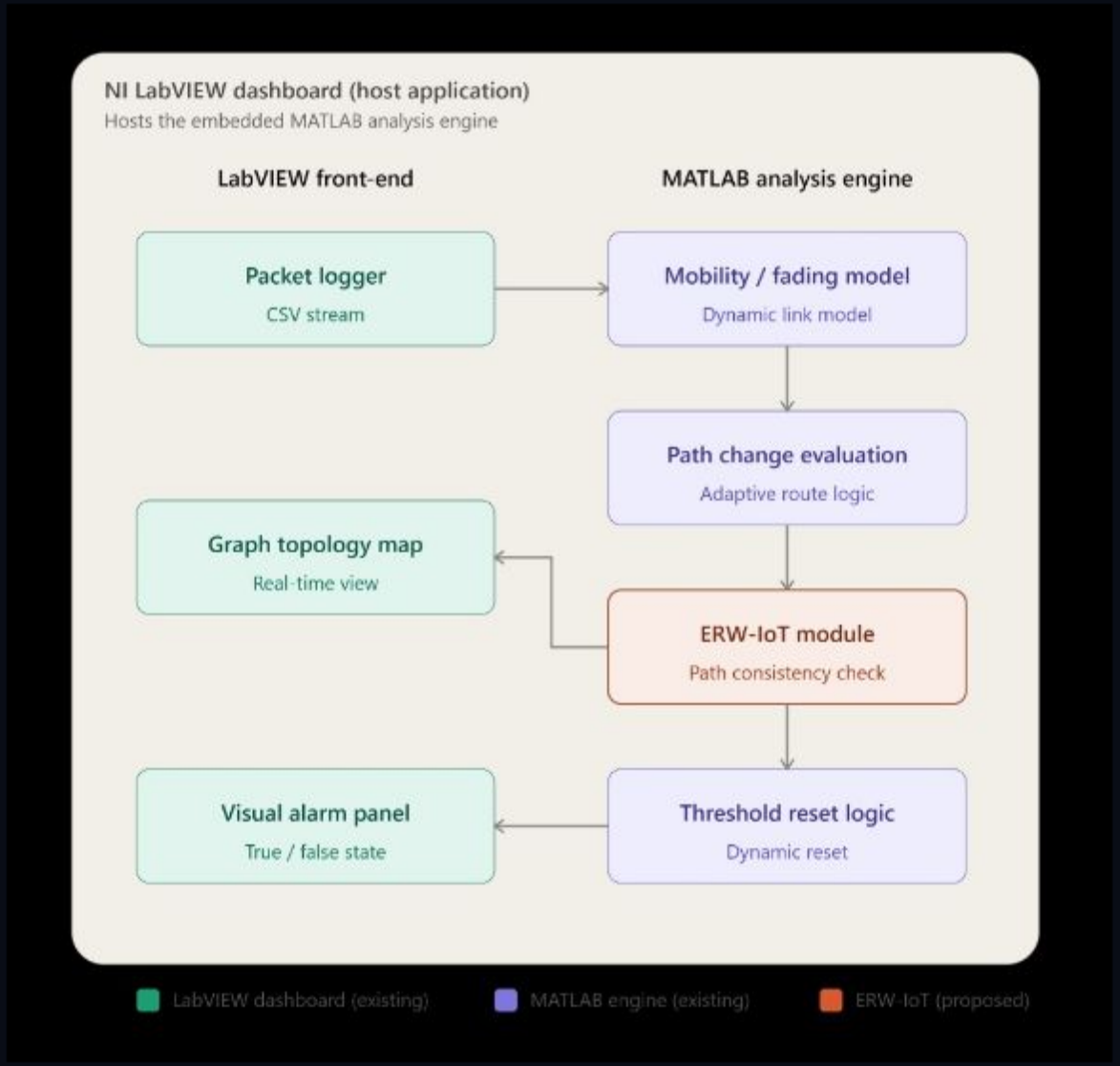


Overall Block Diagram

LabVIEW Front-End Host: Captures live CSV telemetry via Packet Logger, displays real-time topology map, and controls visual alarm state.

MATLAB Engine Brain: Runs mobility/fading models, evaluates candidate path changes, and updates dynamic threshold reset logic.

Proposed ERW-IoT Module: Intercepts path reroutes to perform physical consistency checks before updating alert thresholds or topology maps.



Theoretical Background

Computer Networks & Tables

Wireless networks rely on dynamic routing tables containing destination address, next hop, and distance metrics to continuously establish optimal packet forwarding paths across active nodes.

Network Latency Bounds

Includes propagation delay, queueing delay, and processing time. In legitimate hop-by-hop wireless links, cumulative latency scales proportionally with the physical distance and hop count.

DODAG in RPL / IoT

Destination-Oriented Directed Acyclic Graph structured by RPL protocol for low-power and lossy networks (6LoWPAN). Nodes dynamically select preferred parent nodes based on rank and link cost.

Wormhole Attacks in IoT

Adversaries tunnel packets between distant network areas through an out-of-band link, creating an illusion of a short high-quality route to disrupt routing trees and intercept sensitive traffic.

System Processing Methodology

01 LABVIEW

Packet Logging & Telemetry

The Packet Logger records continuous raw wireless telemetry, formatting network events into a CSV stream passed into the processing pipeline.

02 MATLAB

Fading & Path Evaluation

Mobility model calculates link fading dynamics as nodes move, triggering Path Change Evaluation to suggest updated optimal routes.

03 ERW-IOT

Path Consistency Validation

ERW-IoT checks physical plausibility by comparing expected latency vs. hop count to catch "too good to be true" wormhole shortcuts.

04 DASHBOARD

Feedback & Alarm Trigger

If suspicious, Threshold Reset logic flips the visual alarm to True and highlights the malicious path live on the Graph Topology Map.

Project Module Description

LabVIEW Host Dashboard

Packet Logger (CSV Stream)
Functions as a high-throughput network flight recorder, storing raw telemetry and packet headers for analysis.

Graph Topology Map
Renders live graphical rendering of active node links and flags suspicious wormhole tunnels with high-visibility color overlays.

Visual Alarm Panel
Binary status indicator (True/False state) providing clear and immediate visual feedback for human network monitoring.

MATLAB Analysis Engine & ERW-IoT

Mobility & Fading Model
Simulates dynamic wireless link fading and node mobility patterns in real-time.

ERW-IoT Consistency Check
Core innovation comparing hop count, round-trip delays, and physical limits to isolate wormhole tunnels.

Threshold Reset Logic
Dynamically recalculates network baseline metrics to update system alarm thresholds cleanly.

Project Implementation Timeline

P1

WEEKS 1 – 3

Architecture Design

LabVIEW UI dashboard wireframing, CSV logging stream initialization, and LabVIEW-MATLAB co-simulation API setup.

P2

WEEKS 4 – 7

MATLAB Engine & ERW-IoT

Implementation of mobility/fading models, RPL DODAG simulator, and development of the ERW-IoT path validation algorithms.

P3

WEEKS 8 – 10

Co-Simulation & Attack Injection

Synthetic wormhole tunnel injection, tuning hop latency consistency bounds, and evaluating detection accuracy.

P4

WEEKS 11 – 12

Dashboard Integration

Final integration of topology map highlight overlays, visual alarm panel triggers, dynamic threshold resets, and documentation.

References

- https://www.ni.com/en/shop/labview.html
- https://www.ni.com/docs/en-US/bundle/labview-api-ref/page/functions/script-node.html
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